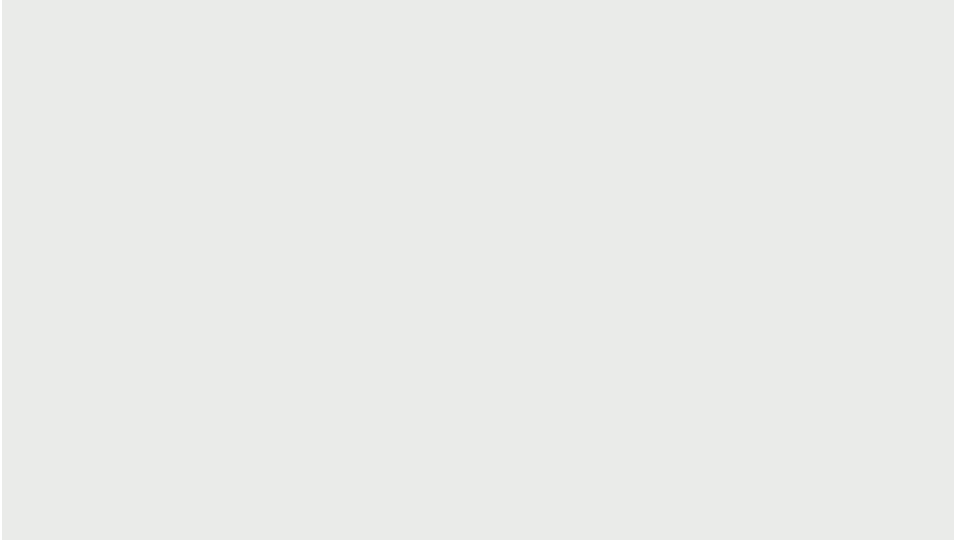
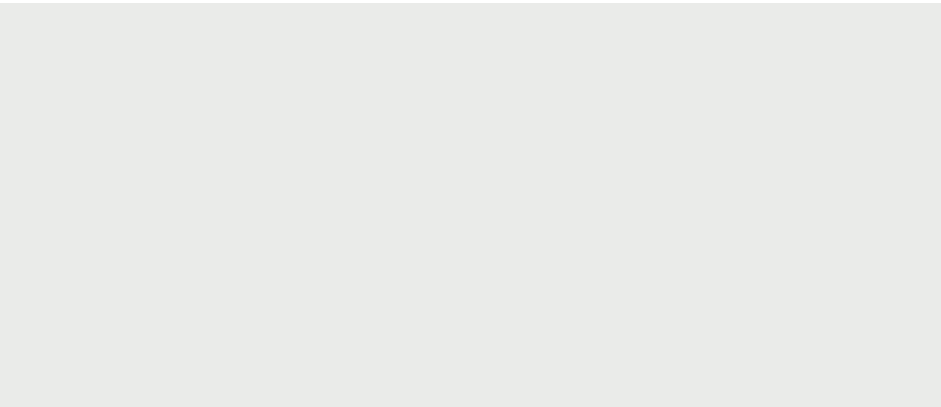




RESEARCH PROPOSAL

Machine Learning for Early Detection of University Dropout

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Problem Statement

Our study focuses on predicting college drop-out rates using academic and socioeconomic factors

Research Question



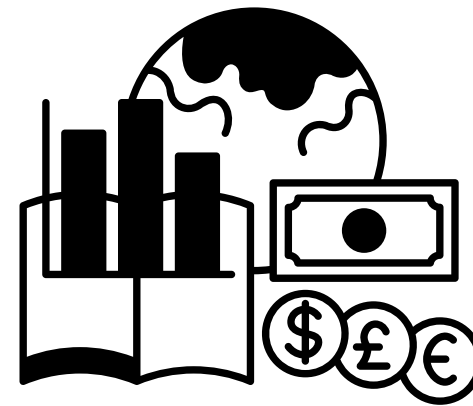
Predict indicators of dropping out to better inform interventions that improve academic outcomes.

Relevance of the Study



By accurately predicting which students are at risk of dropping out, universities can proactively deploy targeted support services (academic tutoring, mental health counseling) before student reaches a breaking point.

Key Variables



Socioeconomic: Parents' career, Tuition status, Inflation GDP, Unemployment Rate at time of enrollment



Demographic: Marital status, gender, age, nationality, displacement/homelessness



Academic: Course, number of credits, enrollment status (daytime/nighttime), grades

Methodology

Feature & Model Design

Task: Multiclass classification
Categories: Dropout, Enrolled, Graduate

Feature Sets

- D₀ (Early) - ex-ante variables (Early warning)
- D₁ (Mid) - D₀ + 1st sem (Intermediate signal)
- D₂ (Full) - D₁ + 2nd sem (Upper bound prediction)

Purpose: consider timing in decision-making

Model Selection

- M₁: Multinomial Logit (interpretable baseline)
 - Ridge/Lasso/ElasticNet regularization
- M₂: Random Forest (nonlinear + robust)
- M₃: XGBoost (high-performance benchmark)

Total: 3 dataset x 3 algorithms = 9 models

Model Training & Evaluation

Training Design

- Data Split: train (80%), test (20%)
- 5-fold CV (training set only)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Evaluation Metrics

- F1 (primary)
- Accuracy (secondary)
- ROC-AUC
- Confusion matrix

$$\text{Precision}_{drop} = \frac{TP}{TP + FP}$$

$$\text{Recall}_{drop} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}$$

Both MSE and classification error might not answer our RQ.

Why Model Set-up?

Model Class	Functional Form	Key Property
Linear	Logistic	Interpretability, parametric structure
Bagging	Random Forest	Variance reduction, robustness
Boosting	XGBoost	Bias reduction, nonlinear interactions

Why not:

KNN – categorical variables, curse of dimensionality
Neural network – Data inefficiency, hyperparameter sensitivity

Proposed Timeline

Weeks 1-2

- Data cleaning and initial analyses
- Logistic regression (regularization)
- Data cleaning- Grayson and Eric
- Initial analyses- Joy and Sam

Week 3

- Secondary analyses
- Gradient boosting (XGBoost), random forest?
- Random forest- Grayson and Joy
- Gradient boosting- Eric and Sam

Week 4

- Identify the analyses that best answer our question
- Begin crafting presentation
- Collaborative, each person contributes to discussion/presentation

Week 5

- Finalize presentation
- Rehearse materials and identify any last minute errors
- Collaborative